

JEE MAIN REVISION MASTER: LIQUID SOLUTION

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Raoult's Law & Vapor Pressure (Ideal & Non-ideal)	21 PYQs	Vapor Pressure
High	Colligative Properties & Van 't Hoff Factor	11 PYQs	Colligative

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Raoult's Law & Vapor Pressure (Ideal & Non-ideal)	$P_{\text{total}} = P_A^{\text{circ}} x_A + P_B^{\text{circ}} x_B, \quad y_A = \frac{P_A}{P_{\text{total}}} = \frac{P_A^{\text{circ}} x_A}{P_{\text{total}}}, \quad P = K_H \cdot x \text{ (Henry)}$
Colligative Properties & Van 't Hoff Factor	$\frac{P^{\text{circ}} - P}{P^{\text{circ}}} = i \cdot x_B, \quad \Delta T_b = i K_b m, \quad \Delta T_f = i K_f m, \quad \Pi = i C R T, \quad i = 1 + (n - 1)\alpha$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Raoult's Law & Vapor Pressure (Ideal & Non-ideal) (Very High Yield — 21 PYQs)

Core Principle: Partial pressures of volatile mixtures, Dalton's law in vapor phase (y_A), and thermodynamic criteria for deviations ($\Delta H_{\text{(mix)}}$, $\Delta V_{\text{(mix)}}$).

■ **Caution / JEE Pitfall:** Mixture of Chloroform and Acetone exhibits **NEGATIVE** deviation from Raoult's law ($\Delta H_{\text{(mix)}} < 0$, $\Delta V_{\text{(mix)}} < 0$) due to intermolecular hydrogen bonding.

■ **Speed Trick:** Higher Henry's law constant K_H at a given pressure signifies **LOWER** solubility of the gas in liquid.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The relative lowering of vapour pressure produced by dissolving 18 g of urea (Molar mass = 60 g mol⁻¹) in 100 g of water is..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "Match List-I (Laws) with the List-II (Mathematical expressions): List-I List-II (a) Henry's law (i) $p_1 = p_1^0 x_1$ (b) Raoult's law ..." **[Ans: N/A]**
- **[2025 Shift] Q#2 (MCQ):** "Variation of solubility with temperature t for a gas in liquid is shown by the following graphs. The correct representation is..." **[Ans: N/A]**

3.2 Colligative Properties & Van 't Hoff Factor (High Yield — 11 PYQs)

Core Principle: Physical properties depending solely on solute particle count, cryoscopic/ebullioscopic constants, and Van 't Hoff factor i .

■ **Caution / JEE Pitfall:** When determining relative boiling/freezing points of salt solutions, compare effective molality $i \times m$, NOT just nominal molality m !

■ **Speed Trick:** For dimerization/association of n molecules: $i = 1 - \left(1 - \frac{1}{n}\right)\alpha$ implies $i < 1$ (e.g. benzoic acid in benzene, $i \approx 0.5$).

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "When 0.0106 mole of acetic acid was dissolved in 1 kg of water, the freezing point depression for this strength of acid was 0.0205 K. ..." **[Ans: D]**
- **[2026 Shift] Q#4 (MCQ):** "Which of the following is CORRECT with respect to the property mentioned against it?..." **[Ans: D]**
- **[2025 Shift] Q#1 (MCQ):** "Among the following 0.1 m aqueous solutions, which one will exhibit the lowest boiling point elevation, assuming complete ionization of the ..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The relative lowering of vapour pressure produced by dissolving 18 g of urea (Molar mass = 60 g mol⁻¹) in 100 g of water is

- (A) 0.025
- (B) 0.512
- (C) 0.051
- (D) 0.250

Correct Answer: None

Solution Analysis: For a non-volatile solute like urea, the relative lowering of vapour pressure is equal to the mole fraction of the solute. $\frac{p^\circ - p}{p^\circ} = x_{\text{solute}}$ Moles of urea: $n_{\text{urea}} = (18 / 60) = 0.3 \text{ mol}$ Moles of water: $n_{\text{water}} = (100 / 18) = 5.56 \text{ mol}$ Now, mole fraction of urea is $x_{\text{urea}} = \frac{n_{\text{urea}}}{n_{\text{urea}} + n_{\text{water}}} = \frac{0.3}{0.3 + 5.56} = 0.051$ So, the relative lowering of vapour pressure is 0.051 Correct option: Option C.

Benchmark Problem 2: KCET 2026 (Q#2)

When 0.0106 mole of acetic acid was dissolved in 1 kg of water, the freezing point depression for this strength of acid was 0.0205 K. If the calculated freezing point depression is 0.0197 K, Van't Hoff factor (i) and degree of dissociation of acetic acid respectively are

- (A) 0.041 and 1.041
- (B) 1.041 and 0.1041
- (C) 0.041 and 0.041
- (D) 1.041 and 0.041 **[Correct]**

Correct Answer: D

Solution Analysis: For solutions showing abnormal molar mass, Van't Hoff factor is $i = (\text{observed freezing point depression} / \text{calculated freezing point depression})$ Here, $\Delta T_f(\text{observed}) = 0.0205 \text{ K}$ and $\Delta T_f(\text{calculated}) = 0.0197 \text{ K}$ So, $i = (0.0205 / 0.0197) = 1.041$ Acetic acid dissociates in water as $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$ One molecule gives 2 particles on dissociation. For dissociation, $i = 1 + \alpha$ So, $\alpha = i - 1 = 1.041 - 1 = 0.041$ Hence, the Van't Hoff factor is 1.041